Top 50 DAX Formulas in Power BI

📊 1. Basic Aggregation Functions

These are fundamental for summarizing data.

1. **SUM()** – Adds values in a column
2. **AVERAGE()** – Calculates the mean
3. **MIN()** – Returns the smallest value
4. **MAX()** – Returns the largest value
5. **COUNT()** – Counts numeric values
6. **COUNTA()** – Counts non-empty values
7. **COUNTROWS()** – Counts rows in a table
8. **DISTINCTCOUNT()** – Counts unique values
9. **SUMX()** – Iterates and sums expressions
10. **AVERAGEX()** – Iterates and averages expressions

Example:

```
Total Sales = SUM(Sales[Amount])
```

🧮 2. Logical Functions

Used for conditional calculations and decision-making.

11. **IF()** – Conditional evaluation
12. **SWITCH()** – Multiple conditions
13. **AND()** – Logical AND
14. **OR()** – Logical OR
15. **NOT()** – Logical negation

Example:

```
Profit Status = IF([Profit] > 0, "Profit", "Loss")
```

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📅 3. Date and Time Functions

Essential for time intelligence analysis.

16. **TODAY()** – Returns the current date
17. **NOW()** – Returns current date and time
18. **DATE()** – Creates a date
19. **YEAR()** – Extracts year
20. **MONTH()** – Extracts month
21. **DAY()** – Extracts day
22. **DATEDIFF()** – Calculates date differences
23. **CALENDAR()** – Creates a date table
24. **EOMONTH()** – Returns end of month
25. **WEEKNUM()** – Returns week number

Example:

```
Year = YEAR(Sales[Order Date])
```

📈 4. Time Intelligence Functions

Critical for trend and comparative analysis.

26. **TOTALYTD()** – Year-to-date calculation
27. **TOTALMTD()** – Month-to-date calculation
28. **TOTALQTD()** – Quarter-to-date calculation
29. **SAMEPERIODLASTYEAR()** – Compares with last year
30. **DATESYTD()** – Returns YTD dates
31. **DATESMTD()** – Returns MTD dates
32. **DATESQTD()** – Returns QTD dates
33. **DATEADD()** – Shifts dates
34. **PARALLELPERIOD()** – Returns parallel periods
35. **PREVIOUSMONTH()** – Returns previous month dates

Example:

```
Sales YTD = TOTALYTD(SUM(Sales[Amount]), 'Date'[Date])
```

🔍 5. Filter and Context Functions

These are the backbone of advanced DAX calculations.

36. **CALCULATE()** – Modifies filter context
37. **FILTER()** – Filters a table
38. **ALL()** – Removes filters
39. **ALLEXCEPT()** – Removes filters except specified ones
40. **ALLSELECTED()** – Considers user selections
41. **KEEPFILTERS()** – Preserves existing filters
42. **VALUES()** – Returns distinct values
43. **SELECTEDVALUE()** – Returns a single selected value
44. **HASONEVALUE()** – Checks if only one value exists
45. **ISFILTERED()** – Checks if a column is filtered

Example:

```
Sales All Regions =
CALCULATE(SUM(Sales[Amount]), ALL(Sales[Region]))
```

🔗 6. Relationship and Lookup Functions

Useful for working with multiple tables.

46. **RELATED()** – Fetches values from related tables
47. **RELATEDTABLE()** – Returns related rows
48. **LOOKUPVALUE()** – Finds values without relationships
49. **USERELATIONSHIP()** – Activates inactive relationships
50. **CROSSFILTER()** – Modifies cross-filter direction

Example:

```
Product Category = RELATED(Products[Category])
```

🎯 Bonus Tips for Mastering DAX

- **Start with CALCULATE()**—it's the heart of DAX.
- **Understand row context vs. filter context** for advanced analytics.
- **Always use a proper Date Table** for time intelligence functions.
- **Practice with real datasets** to build expertise.
- **Create measures instead of calculated columns** whenever possible for better performance.